Xi'an Jiaotong-Liverpool University
西交利物浦大學

# Module Code: CHE105
# Module Title: Introductory Organic Chemistry for Non-Chemists
# Semester 1, 2021/22 Year 2

# RESIT EXAMINATIONS

## Duration: 2 hours

**Module Examiner: Dr. E.Amigues** **Department: Chem. Tel: 0512 8816 7123**

Instructions to Candidates:

1. Answer all questions.

2. The maximum mark is 100.

3. You can answer the questions in any order you prefer, e.g. start from Q4.

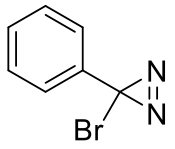
4. Answers should be written on the Answer Booklet provided; the left-hand page of the Answer Booklet can be used for drafting work when necessary.

5. Answers should be written using pen, black or blue, **NO PENCIL**.

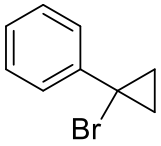
6. Only solutions written in English are accepted.

**Question 1 (8 marks)**

**a.** It is found that bromodiazirine **1** undergoes loss of bromine to produce a cation much more easily than bromocyclopropane **2**.

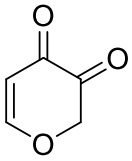



**1**

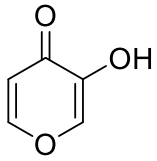



**2**

**b.** Compound **3** is found almost entirely in its enol form **4**. Why?

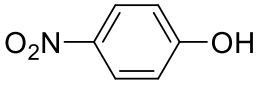



**3**

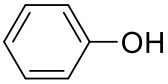



**4**

**c.** *p*-Nitrophenol has p*K*a≈ 8 whereas phenol has p*K*a ≈10




**5**



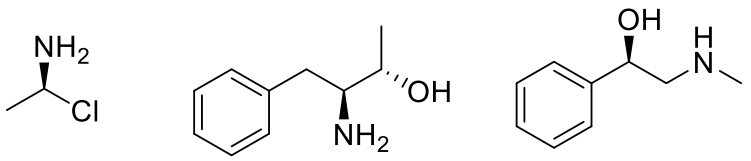

**6**

**d.** The pKa of the doubly allylic proton of cyclopentadiene is ≈ 16, whilst the pKa of the doubly allylic proton of 1,4-pentadiene is ≈ 40.

**Question 2 (12 marks)**

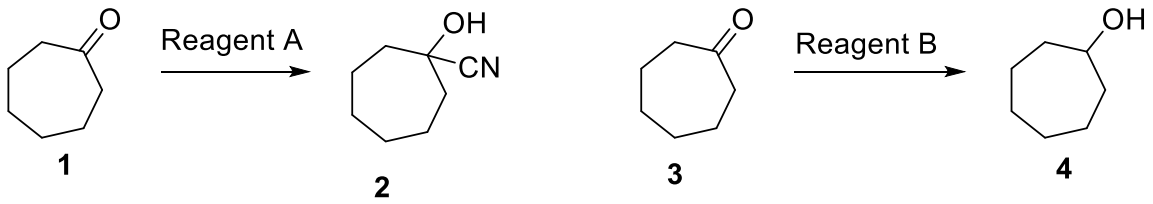
Assign the configuration (*R* or *S*) to each stereocentre in the molecules below. Clearly show how you have worked out your answer. (6 marks for the assignment, 6 marks for the justification).

Show the CIP priority rules for every single stereocenter




**Question 3 [10 marks]:**

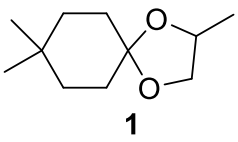
Propose suitable reagents and provide plausible mechanisms for the formation of compounds **2** and **4** (5 marks for each reaction, 2 for the reagents, 3 for the mechanism).



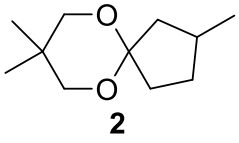

**1** **2** **3** **4**

**Question 4 (10 marks)**

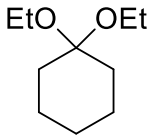
Which compounds were used to produce each of the following acetals **1** to **4**.



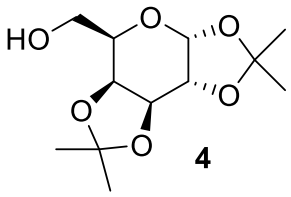

**1**




**2**



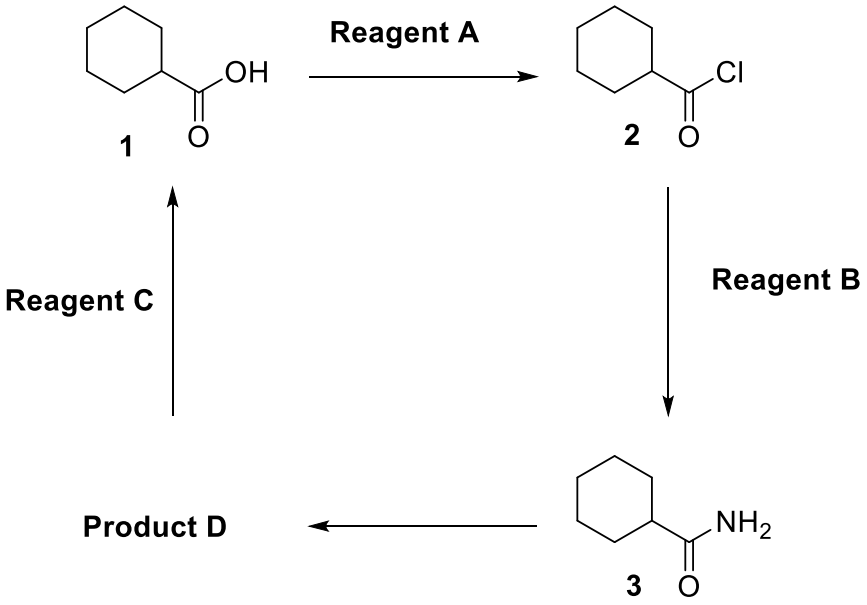

**3**




**4**

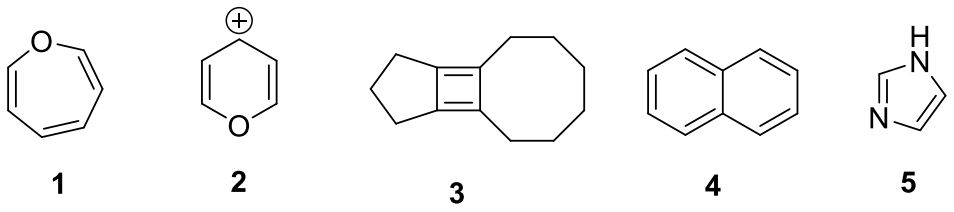
**Question 5 (10 marks)**

Identify the reagents **A, B, C** and the product **D**. No mechanisms required. (2.5 marks each)



**Question 6 (10 marks)**

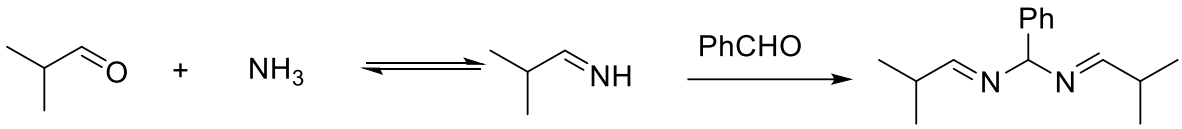
Which structural features does a compound need to have in order to be aromatic? (5 marks) Which of the following compounds are aromatic? Fully justify your answer. (5 compounds, 1 mark each).




**1** **2** **3** **4** **5**

**Question 7 (20 marks)**

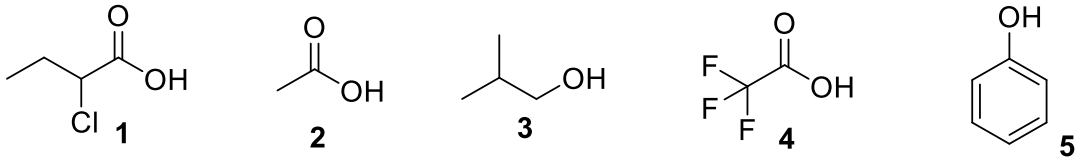
Suggest mechanisms for these reactions that lead to the formation of product **A**. (10 marks)




**A**

**Question 8 (10 marks)**

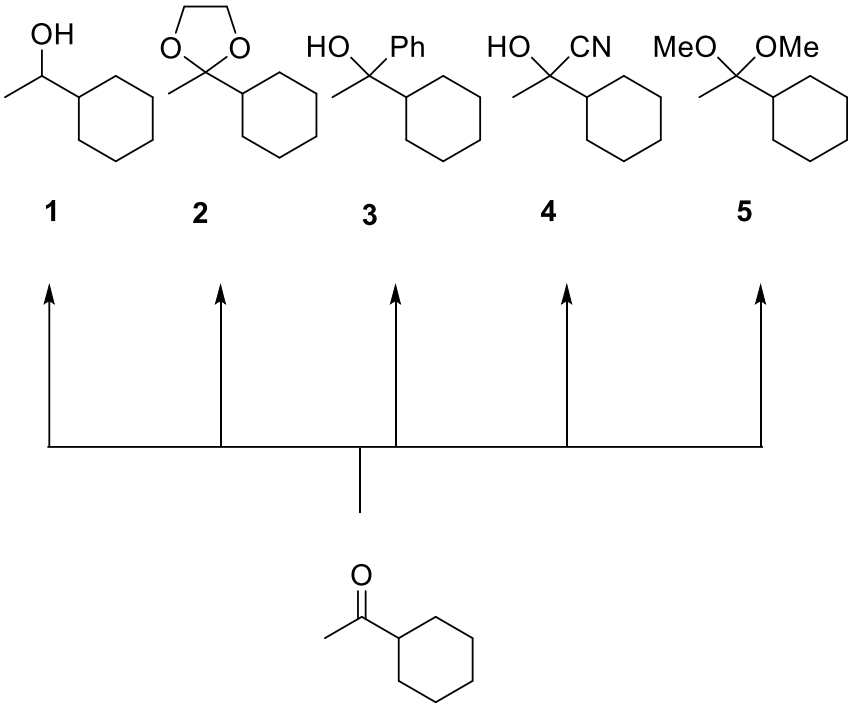
Rank compounds **1** to **5** in order of increasing acidity. Explain your answers.




**1** **2** **3** **4** **5**

**Question 9 (10 marks)**

Identify suitable reagents to convert cyclohexylmethylketone into **1**, **2**, **3, 4** and **5**. No mechanisms required. (2 marks each).




*The End of the Paper*